

Lab Report: EEE 4878

Group No. 10 — Lab No. 05

LAB TITLE

Layout Design and Verification of a CMOS Inverter Using Cadence Virtuoso Layout Suite XL

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Abstract

This experiment presents the complete custom CMOS integrated-circuit design flow using Cadence Virtuoso. The principal objective was to understand how a transistor-level schematic is converted into a physical layout suitable for semiconductor fabrication.

A CMOS inverter was designed using a 45 nm CMOS technology library. The inverter consists of a PMOS transistor in the pull-up network and an NMOS transistor in the pull-down network. To compensate for the lower hole mobility in the PMOS device, its width was selected to be twice that of the NMOS device, supporting more balanced switching characteristics.

The design process included schematic creation and verification, layout generation, transistor placement, multi-finger configuration, routing, body-tie implementation, and physical verification. Design Rule Checking (DRC) was used to assess manufacturability, while Layout Versus Schematic (LVS) verification confirmed that the physical layout matched the original schematic.

Keywords: CMOS inverter, Cadence Virtuoso, layout design, 45 nm CMOS, DRC, LVS, multi-finger MOSFET.

Chapter 1

Introduction

Integrated-circuit design consists of several stages, beginning with circuit description and ending with physical implementation. A schematic represents the electrical behavior and connectivity of a circuit, whereas a layout represents the geometric structures that will be fabricated on silicon.

The CMOS inverter is a fundamental building block of digital VLSI systems. It uses complementary MOS devices to provide logic inversion with low static power consumption and high noise immunity.

In this experiment, a schematic-driven layout methodology was followed. Cadence Virtuoso Layout Suite XL was used to generate the physical layout from the schematic. The layout was then routed and evaluated using standard physical verification techniques.

Chapter 2

Theory

2.1 CMOS Inverter Operation

A CMOS inverter consists of two complementary MOS transistors. The PMOS device is connected between the supply voltage and the output node, while the NMOS device is connected between the output node and ground.

When the input voltage is low:

- the PMOS transistor turns on;
- the NMOS transistor turns off; and
- the output becomes logic high.

When the input voltage is high:

- the NMOS transistor turns on;
- the PMOS transistor turns off; and
- the output becomes logic low.

2.2 Transistor Sizing

For a fixed process and operating condition, the drain-current capability is approximately proportional to the transistor aspect ratio:

$$I_D \propto \frac{W}{L}. \quad (2.1)$$

NMOS devices have higher carrier mobility than PMOS devices. Consequently, a larger PMOS width is commonly used to obtain a drive strength comparable to that of the NMOS transistor. Table 2.1 lists the dimensions used in this experiment.

Table 2.1: CMOS inverter transistor dimensions.

Device	Width, W	Length, L
PMOS	960 nm	45 nm
NMOS	480 nm	45 nm

Chapter 3

Tools Used

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Suite XL
- 45 nm CMOS process design kit (PDK)
- Pegasus DRC verification tool
- Pegasus LVS verification tool
- Quantus parasitic extraction tool

Chapter 4

Procedure

The layout was designed and verified using the following procedure:

1. A working library was created and attached to the required technology library.
2. The CMOS inverter schematic was created in Cadence Virtuoso.
3. PMOS and NMOS devices were placed from the 45 nm technology library.
4. The PMOS width was set to 960 nm, the NMOS width to 480 nm, and both channel lengths to 45 nm.
5. The PMOS source was connected to `avdd`, the NMOS source to `avss`, both gates to `vin`, and both drains to `vout`.
6. An initial schematic-driven layout was generated using Layout XL.
7. The transistors were configured using multi-finger structures and were placed to support compact routing.
8. Body ties were added, and all signal and supply nets were routed.
9. DRC was performed to identify fabrication-rule violations.
10. LVS verification was performed to compare the extracted layout with the schematic.

Chapter 5

Results and Observations

5.1 CMOS Inverter Schematic

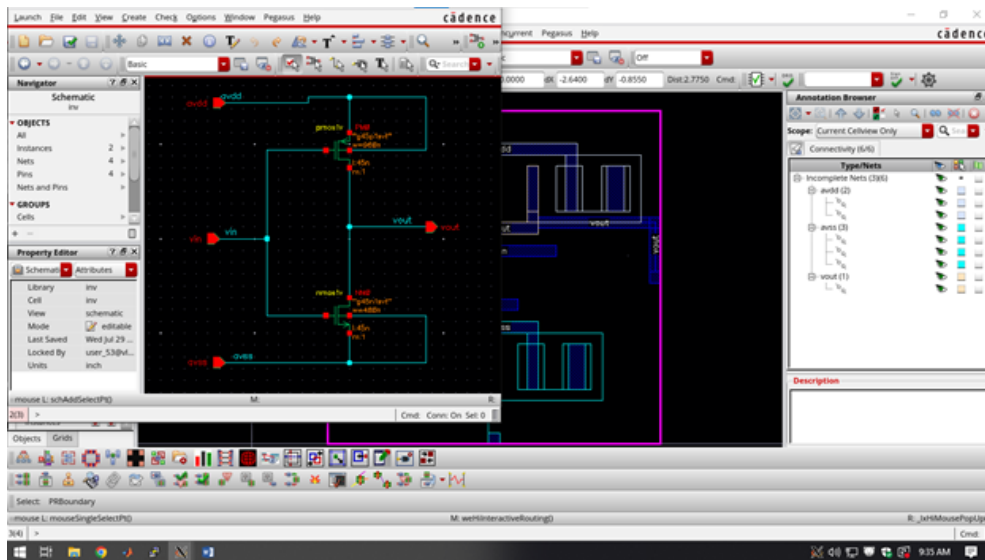


Figure 5.1: CMOS inverter schematic created in Cadence Virtuoso.

Figure 5.1 shows the complementary connection of the PMOS and NMOS devices. The input is applied to both gates, and the output is obtained from the common drain connection.

5.2 Initial Layout Generation

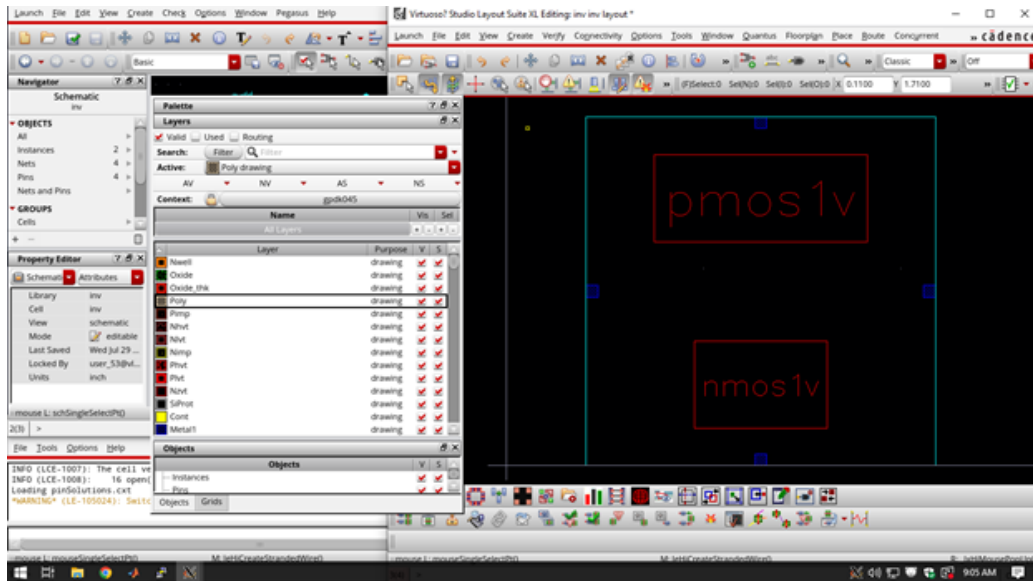


Figure 5.2: Initial layout generated from the schematic.

The generated layout contains the physical representations of the PMOS and NMOS devices. At this stage, the routing between terminals was incomplete.

5.3 Multi-Finger Transistor Layout

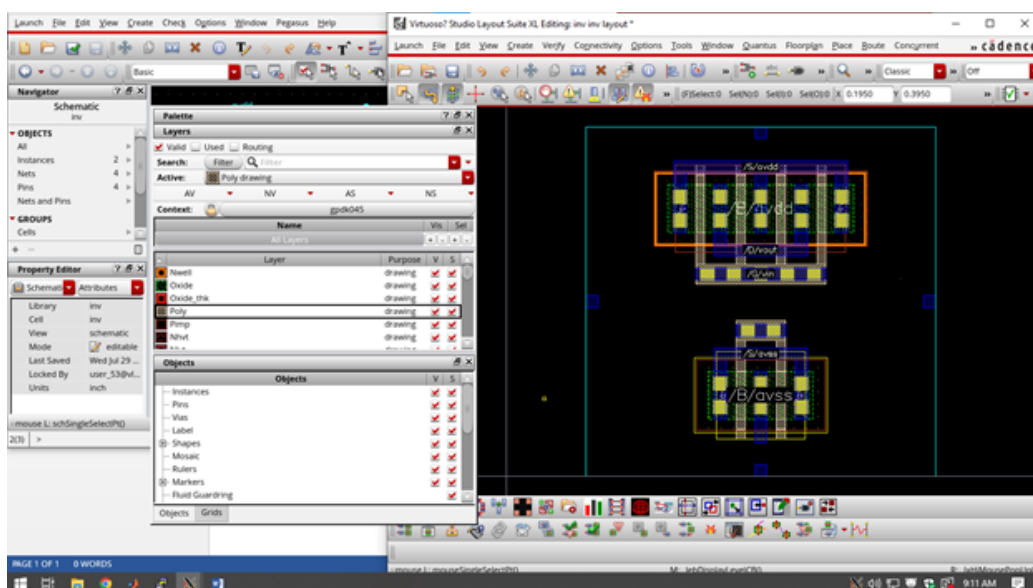


Figure 5.3: Multi-finger transistor structure.

The wide transistor was divided into multiple smaller fingers. This arrangement can reduce gate resistance, improve routing flexibility, and support a more compact layout while satisfying the applicable design rules.

5.4 Body-Tie Connection

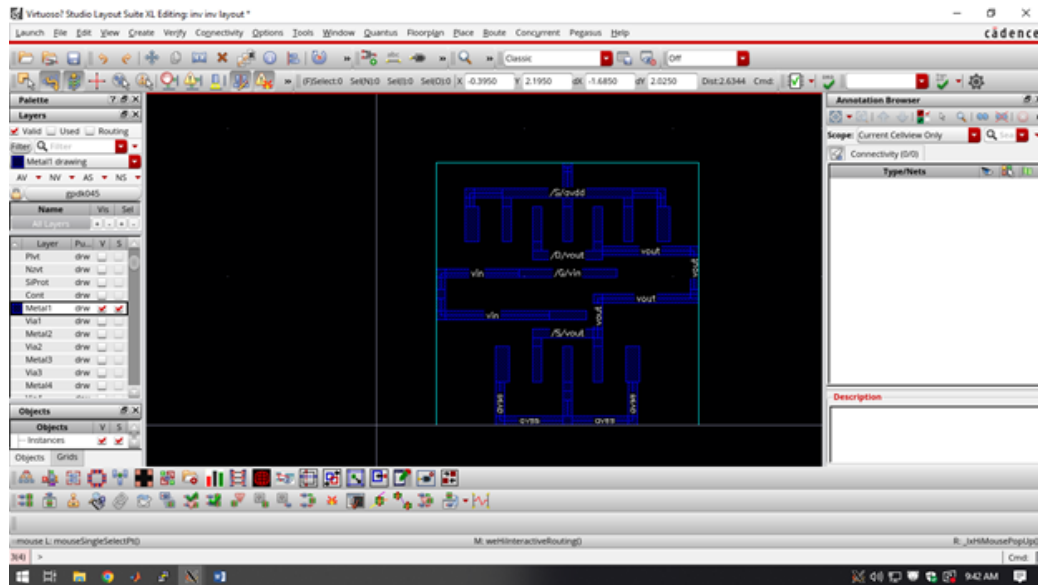


Figure 5.4: Body-tie implementation.

The PMOS body was connected to **avdd**, and the NMOS body was connected to **avss**. These connections establish defined body potentials and help prevent unintended body-bias effects.

5.5 Completed Layout

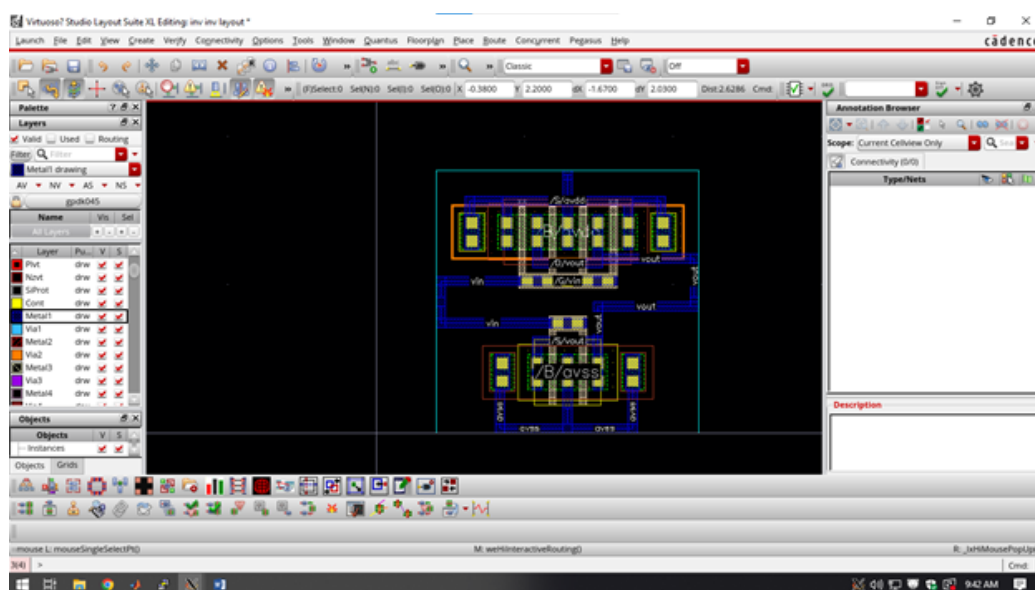


Figure 5.5: Fully routed CMOS inverter layout.

Figure 5.5 shows the completed layout after device placement, body-tie insertion, and routing of the signal and supply nets.

5.6 DRC Verification

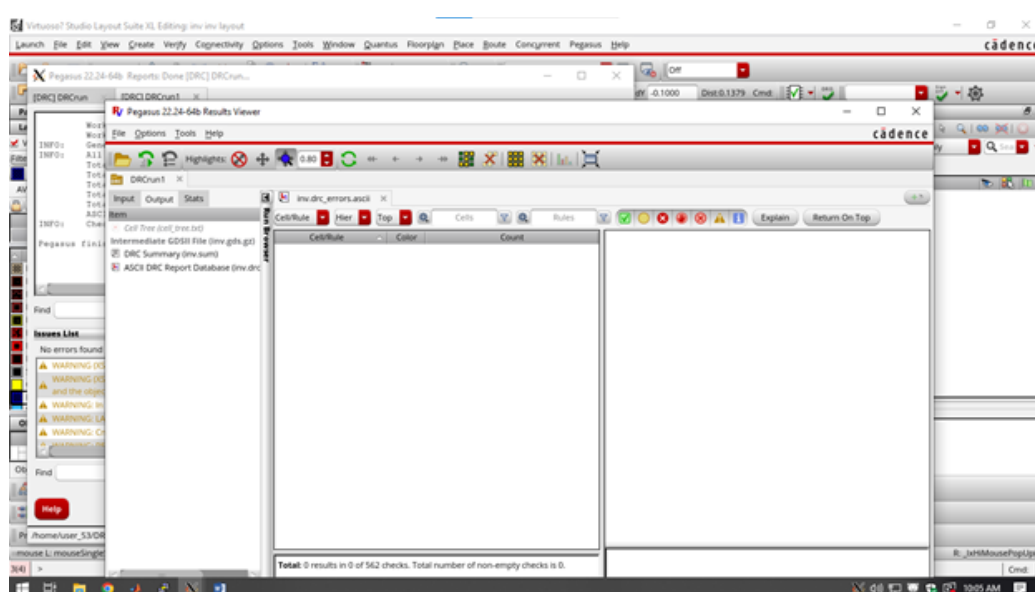


Figure 5.6: DRC verification result.

DRC evaluates geometric constraints such as minimum width, spacing, and enclosure. The reported clean result indicates that no checked design-rule violations remained in the final layout.

5.7 LVS Verification

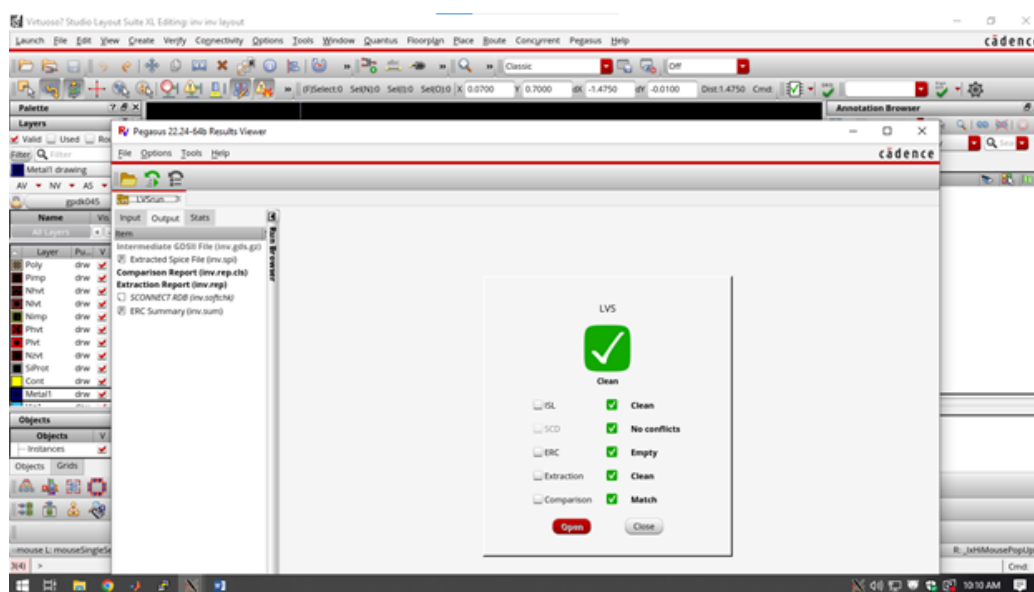


Figure 5.7: LVS verification result.

LVS compares the netlist extracted from the layout with the schematic netlist. The successful comparison confirms that the physical layout implements the intended circuit connectivity.

Chapter 6

Discussion

This experiment demonstrated the transition from circuit design to physical implementation. Schematic-driven layout generation preserved the relationship between electrical devices and their corresponding physical structures, while manual placement and routing were used to produce a compact implementation.

The multi-finger structures divided each wide device into narrower parallel segments. This improved placement and routing flexibility and can reduce gate resistance. Body connections were equally important: tying the PMOS body to `avdd` and the NMOS body to `avss` established the intended body potentials and avoided floating-body behavior.

DRC and LVS addressed complementary verification requirements. DRC checked the layout geometry against fabrication constraints, whereas LVS checked electrical equivalence between the layout and schematic. Passing both checks provided evidence that the final design was physically valid and correctly connected.

Chapter 7

Conclusion

The CMOS inverter was successfully designed, laid out, routed, and verified using Cadence Virtuoso. The experiment provided practical experience with transistor sizing, schematic-driven layout generation, multi-finger device implementation, routing, body ties, and physical verification. The reported clean DRC and successful LVS comparison confirmed that the final layout met the checked physical design rules and matched the intended schematic connectivity.

Bibliography

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